

9 Continuous Random Variables: Probability Density Functions

- Simulated values of a continuous random variable are usually plotted in a **histogram** which groups the observed values into “bins” and plots densities or frequencies for each bin. (R: `hist`).
- In a histogram *areas* of bars represent relative frequencies; the axis which represents the height of the bars is called “density” (R: `hist` with `freq = FALSE`).
- The distribution of a continuous random variable can be described by a **probability density function (pdf)**, for which *areas under the density curve determine probabilities*.

9.1 Uniform Distributions

Example 9.1. In the meeting problem assume that Regina’s arrival time X (minutes after noon) follows a $\text{Uniform}(0, 60)$ distribution. Figure 9.1 displays a histogram of 10000 simulated values of X .

1. Sketch a spinner to simulate values of X .
2. Sketch a plot of the pdf of X .

Repetition	X
1	49.947381
2	9.921521
3	1.848228
4	35.695739
5	58.473455
6	10.768278

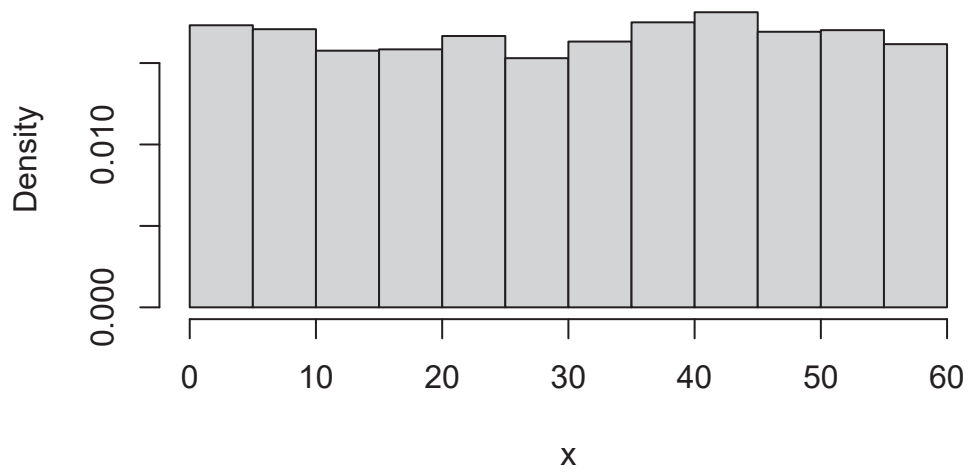


Figure 9.1: Simulated values from a $\text{Uniform}(0, 60)$ distribution

3. Use the pdf to find the probability that Regina arrives before 12:15.
4. Use the pdf to find the probability that Regina arrives after 12:45.
5. Use the pdf to find the probability that Regina arrives between 12:15:00 and 12:16:00.
6. Use the pdf to find the probability that Regina arrives between 12:15:00 and 12:15:01.

7. Compute $P(X = 15)$, the probability that Regina arrives at the exact time 12:15:00 (with infinite precision).

- **The probability that a continuous random variable X equals any particular value is 0.** That is, if X is continuous then $P(X = x) = 0$ for all x .
- Even though any specific value of a continuous random variable has probability 0, *intervals* still can have positive probability.
- In practical applications involving continuous random variables, “equal to” really means “close to”, and “close to” probabilities correspond to intervals which can have positive probability.
- A continuous random variable X has a **Uniform distribution** with parameters a and b , with $a < b$, if its probability density function f_X satisfies

$$f_X(x) \propto \text{constant}, \quad a < x < b \quad (9.1)$$

$$= \frac{1}{b-a}, \quad a < x < b. \quad (9.2)$$

- If X has a $\text{Uniform}(a, b)$ distribution then

$$E(X) = \frac{a+b}{2} \quad (9.3)$$

$$\text{Var}(X) = \frac{|b-a|^2}{12} \quad (9.4)$$

$$\text{SD}(X) = \frac{|b-a|}{\sqrt{12}} \quad (9.5)$$

9.2 Probability Density Functions

- The continuous analog of a probability mass function (pmf) is a *probability density function* (pdf).
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A pdf of a continuous random variable must be *integrated* to find probabilities of related events.

- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx, \quad \text{for all } -\infty \leq a \leq b \leq \infty \quad (9.6)$$

- For a continuous random variable X with pdf f_X , the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \geq 0 \quad \text{for all } x, \quad (9.7)$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \quad (9.8)$$

Example 9.2. Continuing Example 9.1, we will now we assume Regina's arrival time has pdf

$$f_X(x) = \begin{cases} cx, & 0 \leq x \leq 60, \\ 0, & \text{otherwise.} \end{cases}$$

where c is an appropriate constant.

1. Sketch a plot of the pdf. Generally, what does this say about Regina's arrival time?
2. Find the value of c and specify the pdf of X .
3. Find the probability that Regina arrives before 12:15.
4. Find the probability that Regina arrives after 12:45. How does this compare to the previous part? What does that say about Regina's arrival time?

5. Find $P(X = 15)$, the probability that Regina arrives at the exact time 12:15 (with infinite precision).
 6. Find $P(X = 45)$, the probability that Regina arrives at the exact time 12:45 (with infinite precision).
 7. Find the probability that Regina arrives between 12:15 and 12:16.
 8. Find the probability that Regina arrives between 12:45 and 12:46. How does this compare to the probability for 12:15 to 12:16? What does that say about Regina's arrival time?
- **The probability that a continuous random variable X equals any particular value is 0.** That is, if X is continuous then $P(X = x) = 0$ for all x .
 - For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value. However, we can consider the probability that a random variable is *close to* a particular value.
 - The density $f_X(x)$ at value x is *not* a probability.
 - Rather, the density $f_X(x)$ at value x is related to the probability that the RV X takes a value "close to x " in the following sense

$$P\left(x - \frac{\epsilon}{2} \leq X \leq x + \frac{\epsilon}{2}\right) \approx f_X(x)\epsilon, \quad \text{for small } \epsilon$$

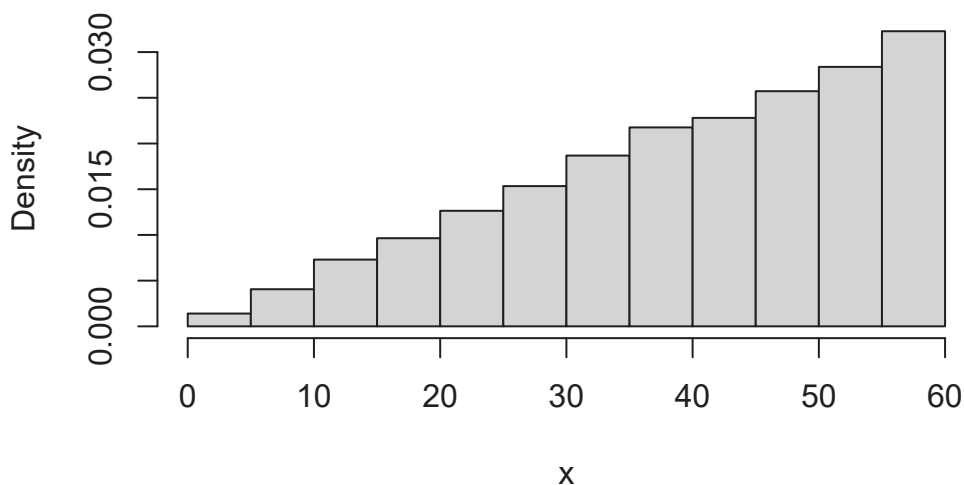


Figure 9.2: Simulated values of X with the pdf in Example 9.2. (We’ll see how this simulation works soon.)

- The quantity ϵ is a small number that represents the desired degree of precision. For example, rounding to two decimal places corresponds to $\epsilon = 0.01$.
- What’s important about a pdf is *relative* heights. For example, if $f_X(x_2) = 2f_X(x_1)$ then X is roughly “twice as likely to be near x_2 than to be near x_1 ” in the above sense.

$$\frac{f_X(x_2)}{f_X(x_1)} = \frac{f_X(x_2)\epsilon}{f_X(x_1)\epsilon} \approx \frac{\mathbb{P}(x_2 - \frac{\epsilon}{2} \leq X \leq x_2 + \frac{\epsilon}{2})}{\mathbb{P}(x_1 - \frac{\epsilon}{2} \leq X \leq x_1 + \frac{\epsilon}{2})}$$

9.3 Exponential Distributions

Example 9.3. Suppose that we model the waiting time, measured continuously in hours, from now until the next earthquake (of any magnitude) occurs in southern CA as a continuous random variable X with pdf

$$f_X(x) = 2e^{-2x}, \quad x \geq 0$$

This is the pdf of the “Exponential(2)” distribution.

1. Sketch the pdf of X . What does this tell you about waiting times?

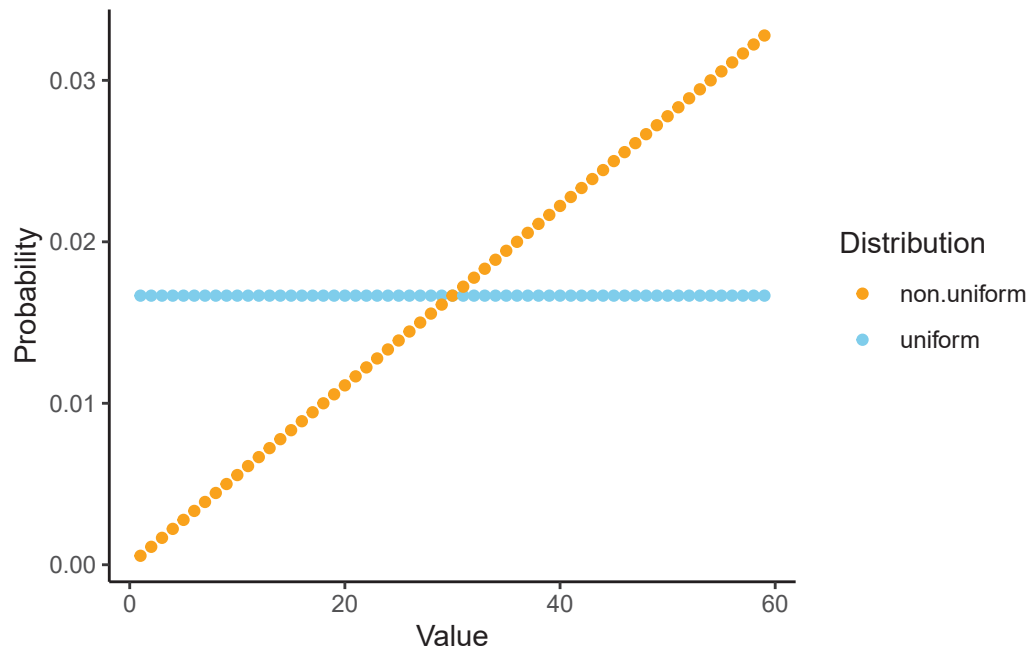


Figure 9.3: Probability of Regina arriving at each minute between noon (0) and 1:00PM (60), to the nearest minute, for the Uniform(0, 60) distribution (blue) and the distribution in Example 9.2 (orange).

2. Without doing any integration, approximate the probability that X rounded to the nearest minute is 0.5 hours.

3. Without doing any integration determine how much more likely that X rounded to the nearest minute is to be 0.5 than 1.0.

4. Compute and interpret $P(X > 0.25)$.

5. Compute and interpret $P(X \leq 3)$.

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- If X has an $\text{Exponential}(\lambda)$ distribution then

$$P(X > x) = e^{-\lambda x}, \quad x \geq 0$$

- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
 - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
 - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)

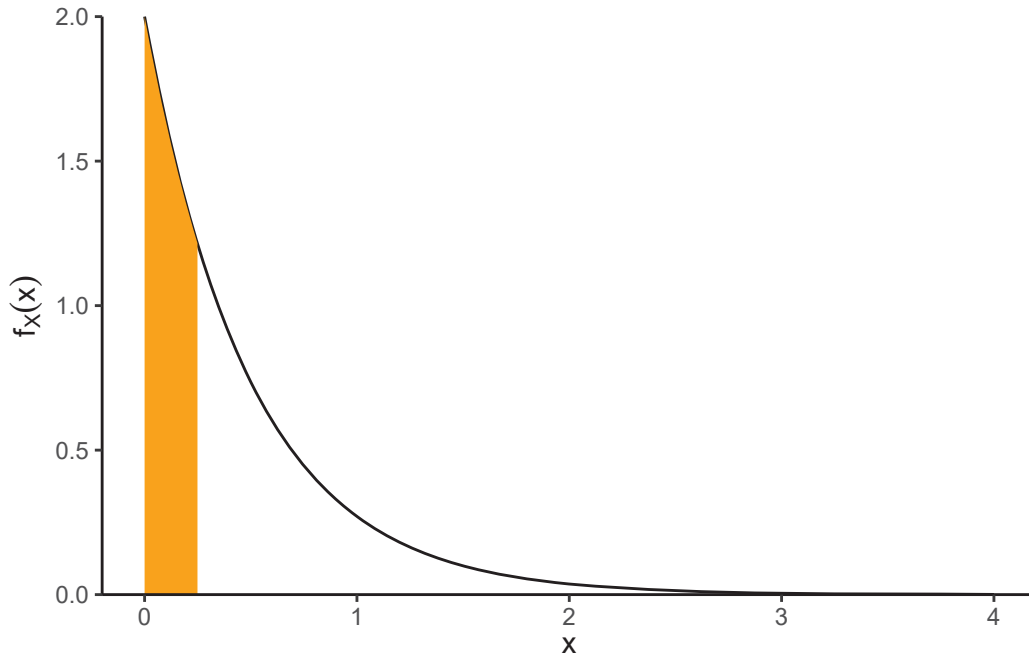


Figure 9.4: Probability density function of X with an Exponential distribution with rate 2. The shaded area represents $P(X < 0.25)$.

9.4 Expected Values

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $E(X)$ representing the probability-weighted average value of X .
- The expected value of a continuous random variable with pdf f_X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $E(X)$ represents *a single number*.
- The expected value is the “balance point” (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $E(\cdot)$ as
 - Simulate lots of values of what’s inside $(\cdot)$
 - Compute the average. This is a “usual” average; just sum all the simulated values and divide by the number of simulated values.

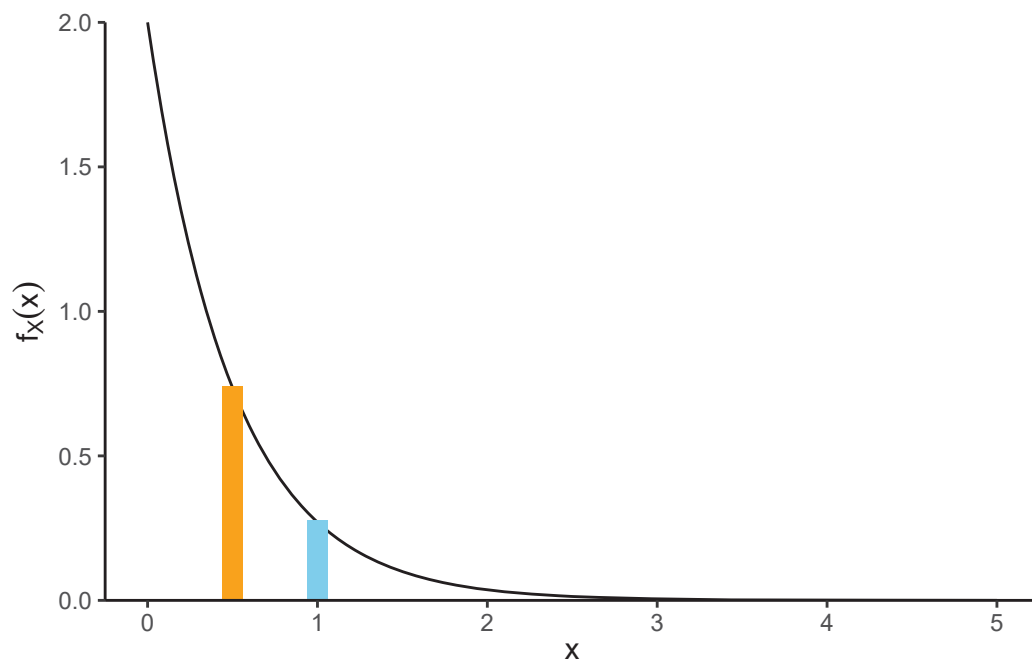


Figure 9.5: Illustration that the probability that X is close to x can be approximated by the area of a rectangle with height equal to the density at x , $f_X(x)$. The density height at $x = 0.5$ is roughly 2.7 times greater than the density height at $x = 1$ so the probability that X is close to 0.5 is roughly 2.7 times greater than the probability that X is close to 1.

Example 9.4. Continuing Example 9.3, where X , the waiting time (hours) from now until the next earthquake (of any magnitude) occurs in southern CA, has an Exponential distribution with rate 2.

1. Use simulation to approximate the long run average value of X .

2. Set up the integral to compute $E(X)$.

3. Interpret $E(X)$.

4. Compute $P(X = E(X))$.

5. Compute $P(X \leq E(X))$. (Is it equal to 50%?)

- If X has an $\text{Exponential}(\lambda)$ distribution then

$$E(X) = \frac{1}{\lambda}$$

- For example, if events happen at rate $\lambda = 2$ per hour, then the average waiting time between events is $1/\lambda = 1/2$ hour.

9.5 Law of the Unconscious Statistician

- The “**law of the unconscious statistician**” (**LOTUS**) says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \quad E[g(X)] = \sum_x g(x)p_X(x) \quad (9.9)$$

$$\text{Continuous } X \text{ with pdf } f_X: \quad E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx \quad (9.10)$$

- LOTUS says we don’t have to first find the distribution of $Y = g(X)$ to find $E[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X) \quad (9.11)$$

- In terms of expected values, in general

$$E(g(X)) \neq g(E(X)) \quad (9.12)$$

The left side $E(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(E(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 9.5. Continuing Example 9.3, where X , the waiting time (hours) from now until the next earthquake (of any magnitude) occurs in southern CA, has an Exponential distribution with rate 2.

1. Use simulation to approximate the variance of X .
2. Set up the integral to compute $E(X^2)$.

3. Compute $\text{Var}(X)$. (The previous part shows $E(X^2) = 0.5$)

4. Compute $\text{SD}(X)$.

- The **variance** of a random variable X is

$$\text{Var}(X) = E\left((X - E(X))^2\right) \quad (9.13)$$

$$= E(X^2) - (E(X))^2 \quad (9.14)$$

- The **standard deviation** of a random variable is

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.
- The definition $E((X - E(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

- That is, variance is the expected value of the square of X minus the square of the expected value of X .
- Variance has many nice theoretical properties. Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.
- If X has an $\text{Exponential}(\lambda)$ distribution then

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

10 Percentiles

Example 10.1. Maggie and Seamus are babies who have just turned one. At their one-year visits to their pediatrician:

- Maggie is 76cm tall and in the 75th percentile of height for girls.
- Seamus is 72cm tall and in the 10th percentile of height for boys.

Explain what these percentiles mean.

- A distribution is characterized by its percentiles.
- Roughly, the value x is the p th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x : $P(X \leq x) = p$.
- Any distribution can be represented by a spinner. A spinner basically describes a distribution by specifying all the percentiles. For example,
 - The 25th percentile goes 25% of the way around the axis (at “3 o’clock”)
 - The 50th percentile goes 50% of the way around the axis (at “6 o’clock”)
 - The 75th percentile goes 75% of the way around the axis (at “9 o’clock”)

10.1 Cumulative Distribution Functions

- Roughly, the value x is the p th percentile of a distribution of a random variable X if p percent of values of the variable are less than or equal to x : $P(X \leq x) = p$.
- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x : x is the (blank) percentile. That is, for an input x , the cdf outputs $P(X \leq x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function, F_X defined by $F_X(x) = P(X \leq x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X .

Example 10.2. Continuing Example 9.2 where Regina's arrival time X (minutes after noon) has pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

Let F_X be the cdf of X .

1. Compute and interpret $F_X(15)$.
2. Compute and interpret $F_X(45)$.
3. Compute and interpret $F_X(45) - F_X(15)$.
4. Compute $F_X(-0.2)$
5. Compute $F_X(70)$
6. Find the cdf of X .

Example 10.3. Let X be the number of heads in 3 flips of a fair coin.

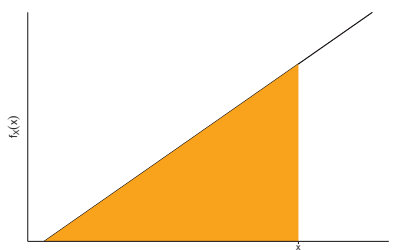


Figure 10.1: PDF

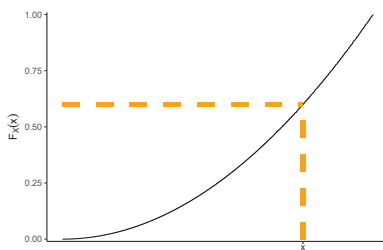


Figure 10.2: CDF

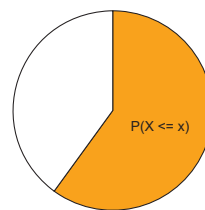


Figure 10.3: Spinner

1. Find the pmf of X and sketch a plot of it.

2. Find the cdf of X and sketch a plot of it.

3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y .

- A cdf is defined for all values of x , regardless if x is a possible value of the RV.
- A cdf is a non-decreasing function: if $x_1 \leq x_2$ then $F_X(x_1) \leq F_X(x_2)$.
- A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- A cdf approaches 1 as the input approaches ∞ : $\lim_{x \rightarrow \infty} F_X(x) = 1$
- The cdf of a *discrete* random variable is a step function.
 - The steps occur at the possible values of the random variable.
 - The height of a particular step corresponds to the probability of that value, given by the pmf.
- The cdf of a *continuous* random variable is a continuous function.
 - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

- The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F'_X = f_X \quad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = P(a < X \leq b)$$

Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.

- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u .
- That is, two random variables have the same distribution if all the percentiles are the same.

10.2 Quantile functions

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in [0, 1]$: the 100 p th percentile is (blank).

Example 10.4. Continuing Example 9.2 where Regina's arrival time X (minutes) had pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

1. Find the median (i.e., 50th percentile) of X .
2. Find the 25th percentile of X .
3. Find the quantile function of X .

4. Construct a spinner for simulating values of X according to its distribution.

- For a *continuous* random variable X with cdf F_X , the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in [0, 1]$: the 100 p th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- The quantile function can be used to create a spinner for a distribution. Basically, the values on the outside boundary of the spinner are scaled based on the quantile function (which is determined by the cdf). Intervals corresponding to regions of higher density (“more likely”) values are stretched out on the spinner boundary; intervals corresponding to regions of lower density (“less likely” values) are shrunk.

Spinner for pdf $f(x) = (2/3600)x$, $0 < x < 60$

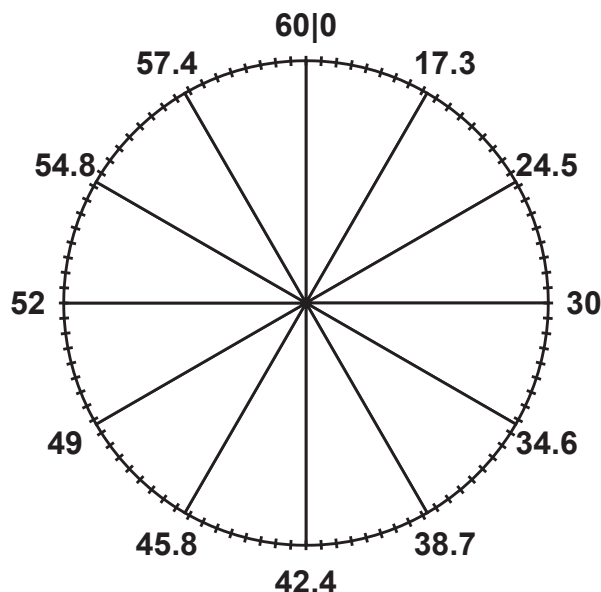


Figure 10.4: Spinner for X in Example 9.2. The axis is transformed according to the quantile function

- **Universality of the Uniform** (or “one spinner to rule them all”). Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F .

- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the $\text{Uniform}(0, 1)$ spinner and transform each resulting value by plugging it into the quantile function.

11 Normal Distributions

- Normal distributions are probably the most important distributions in probability and statistics.
- A continuous random variable X has a **Normal** (a.k.a., Gaussian) distribution with mean $\mu \in (-\infty, \infty)$ and standard deviation $\sigma > 0$ if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty$$

- If X has a $\text{Normal}(\mu, \sigma)$ distribution then

$$\begin{aligned} E(X) &= \mu \\ \text{SD}(X) &= \sigma \end{aligned}$$

- A Normal density is a particular “bell-shaped” curve which is symmetric about its mean μ . The mean μ is a *location* parameter: μ indicates where the center and peak of the distribution is.
- The standard deviation σ is a *scale* parameter: σ indicates the distance from the mean to where the concavity of the density changes. That is, there are inflection points at $\mu \pm \sigma$.

Example 11.1. The pdfs in the plot below represent the distribution of hypothetical test scores in three classes. The test scores in each class follow a Normal distribution. Identify the mean and standard deviation for each class.

- Any Normal distribution follows the “empirical rule” which determines the percentiles that give a Normal distribution its particular bell shape.
- For a Normal distribution:
 - 38% of values are within 0.5 standard deviations of the mean
 - 68% of values are within 1 standard deviation of the mean
 - 87% of values are within 0.5 standard deviations of the mean
 - 95% of values are within 2 standard deviations of the mean
 - 99% of values are within 2.6 standard deviations of the mean
 - 99.7% of values are within 3 standard deviations of the mean
- The table below lists some percentiles of a Normal distribution.

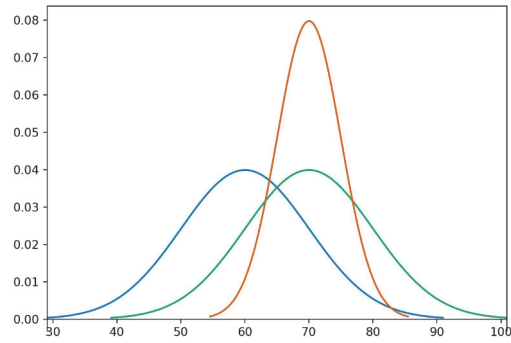


Figure 11.1

Table 11.1: Selected percentiles for a Normal distribution

Percentile	SDs away from the mean
0.1%	3.09 SDs below the mean
0.5%	2.58 SDs below the mean
1%	2.33 SDs below the mean
2.5%	1.96 SDs below the mean
10%	1.28 SDs below the mean
15.9%	1 SDs below the mean
25%	0.67 SDs below the mean
30.9%	0.5 SDs below the mean
50%	0 SDs above the mean
69.1%	0.5 SDs above the mean
75%	0.67 SDs above the mean
84.1%	1 SDs above the mean
90%	1.28 SDs above the mean
97.5%	1.96 SDs above the mean
99%	2.33 SDs above the mean
99.5%	2.58 SDs above the mean
99.9%	3.09 SDs above the mean

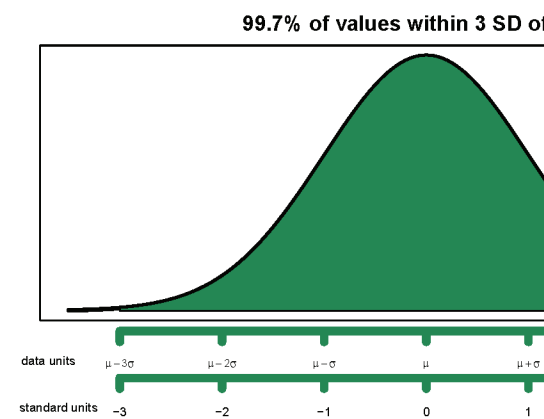
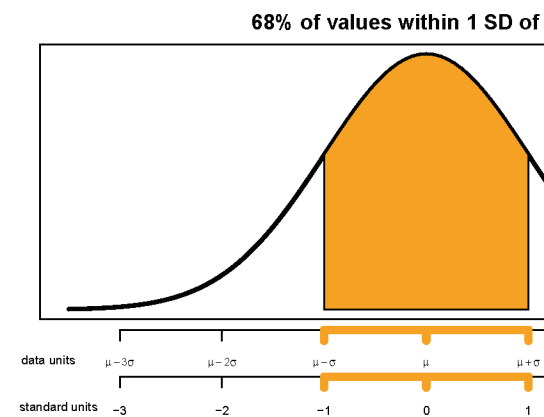
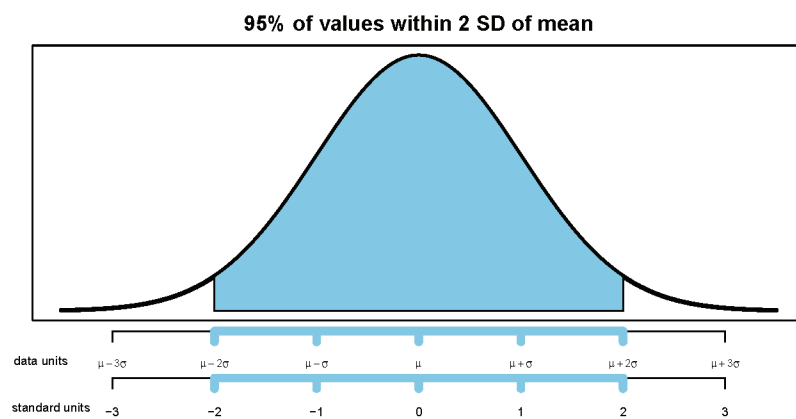
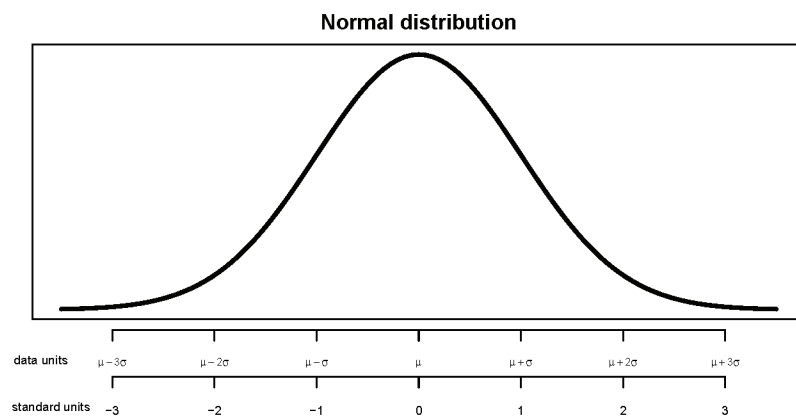


Figure 11.2

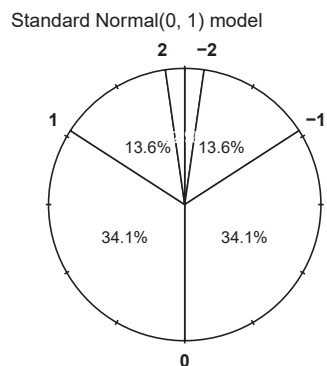
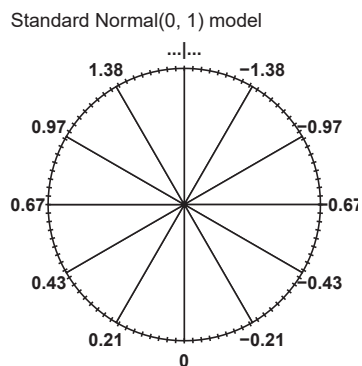


Figure 11.3: A standard Normal(0, 1) spinnerFigure 11.4: A standard Normal(0, 1) spinner.

The same spinner is displayed on both sides, with different features highlighted on the left and right. Only selected rounded values are displayed, but in the idealized model the spinner is infinitely precise so that any real number is a possible outcome. Notice that the values on the axis are *not* evenly spaced.

The same spinner is displayed on both sides, with different features highlighted on the left and right. Only selected rounded values are displayed, but in the idealized model the spinner is infinitely precise so that any real number is a possible outcome. Notice that the values on the axis are *not* evenly spaced.

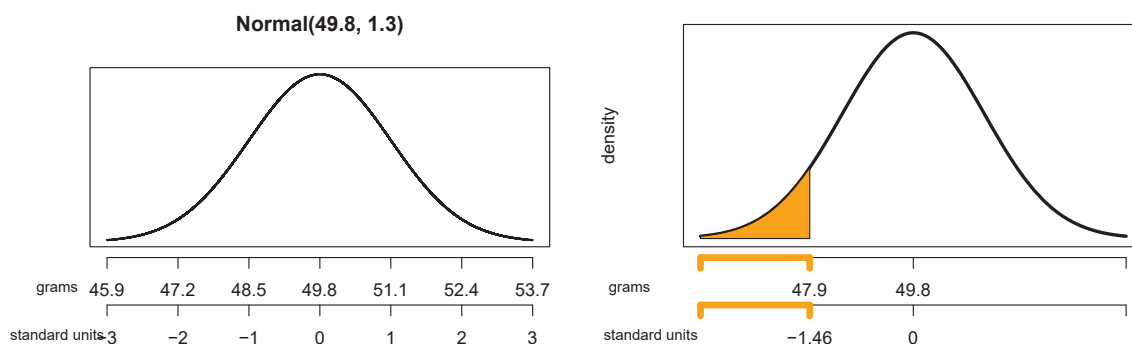
- The Normal(0, 1) distribution (mean 0, SD 1) is called the “Standard Normal” distribution
- Standardization:
 - If X has a Normal(μ, σ) distribution then $Z = \frac{X-\mu}{\sigma}$ has a Normal(0, 1) distribution
 - If Z has a Normal(0, 1) distribution then $X = \mu + \sigma Z$ has a Normal(μ, σ) distribution

Example 11.2. The wrapper of a package of candy lists a weight of 47.9 grams. Naturally, the weights of individual packages vary somewhat. Suppose package weights have an approximate Normal distribution with a mean of 49.8 grams and a standard deviation of 1.3 grams.

1. Sketch the distribution of package weights. Carefully label the variable axis. It is helpful to draw two axes: one in the measurement units of the variable, and one in standardized units.
2. Why wouldn't the company print the mean weight of 49.8 grams as the weight on the package?
3. Estimate the probability that a package weighs less than the printed weight of 47.9 grams.
4. Estimate the probability that a package weighs between 47.9 and 53.0 grams.
5. Suppose that the company only wants 1% of packages to be underweight. Find the weight that must be printed on the packages.

6. Find the 25th percentile (a.k.a., first (lower) quartile) of package weights.

7. Find the 75th percentile (a.k.a., third (upper) quartile) of package weights. How can you use the work you did in the previous part?



Example 11.3. Daily high temperatures (degrees Fahrenheit) in San Luis Obispo in August follow (approximately) A Normal distribution with a mean of 76.9 degrees F. The temperature exceeds 100 degrees Fahrenheit on about 1.5% of August days.

1. What is the standard deviation?

2. Suppose the mean increases by 2 degrees Fahrenheit. On what percentage of August days will the daily high temperature exceed 100 degrees Fahrenheit? (Assume the standard deviation does not change.)

3. A mean of 78.9 is 1.02 times greater than a mean of 76.9. By what (multiplicative) factor has the percentage of 100-degree days increased? What do you notice?

Example 11.4. In a large class, scores on midterm 1 follow (approximately) a $\text{Normal}(\mu_1, \sigma)$ distribution and scores on midterm 2 follow (approximately) a $\text{Normal}(\mu_2, \sigma)$ distribution. Note that the SD σ is the same on both exams. The 40th percentile of midterm 1 scores is equal to the 70th percentile of midterm 2 scores. Compute

$$\frac{\mu_1 - \mu_2}{\sigma}$$

(This is one statistical measure of *effect size*.)

Example 11.5. A bank uses an applicant's score on some criteria to decide whether or not to approve their loan application. Based on past history, the bank has determined that

- Scores for those who repay the loan follow a Normal distribution with mean 60 and SD 8
- Scores for those who do NOT repay the loan follow a Normal distribution with mean 40 and SD 12

When someone applies for a loan the bank does not know whether the applicant will eventually repay the loan. How can the bank use the applicant's score to decide whether or not to approve the loan?

1. Draw sketches of these two normal curves on the same axis.
2. Suggest a decision rule, based on an applicant's score, for deciding whether or not to approve the applicant's loan.
3. Describe the two kinds of classification errors that could be made in this situation.
4. Determine the probability that we incorrectly reject the application of someone who would repay. Shade in the corresponding region under the Normal curve, and interpret this probability.
5. Determine the probability that we incorrectly approve the application of someone who would NOT repay. Shade in the corresponding region under the Normal curve, and interpret this probability.
6. In which direction — smaller or larger — would you need to change the decision rule's cutoff value in order to decrease the probability that an applicant who would repay the loan is incorrectly rejected?
7. Would the probability of the other kind of error — incorrectly approving a loan for an

applicant who would not repay it — increase or decrease with this new cutoff value?

8. Determine the cutoff value needed to decrease the probability that an applicant who would repay the loan is incorrectly rejected to 0.05.
 9. Determine the other error probability with this new cut-off rule.
 10. Now repeat the two previous parts with the goal of decreasing to 0.05 the probability of incorrectly approving an applicant who would not repay the loan.
 11. If you consider the two kinds of errors to be equally serious, how might you decide which of the three decision rules considered so far is the best?
 12. Which error do you think the bank would consider more serious? In which direction would that move the threshold?
- Making decisions/predictions based on data often involves trade-offs.
 - Decreasing the probability of one kind of error often comes at the expense of increasing the probability of another kind of error.

12 Joint Distributions

Example 12.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Recall Table 5.1.

1. Compute and interpret $p_{X,Y}(5,3) = P(X = 5, Y = 3)$.
2. Construct a “flat” table displaying the distribution of (X, Y) pairs, with one pair in each row.
3. Construct a two-way displaying the joint distribution on X and Y .
4. Sketch a plot depicting the joint distribution of X and Y .
5. Starting with the two-way table, how could you obtain $P(X = 5)$?

Table 12.1: Flat table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

(x, y)	$P(X = x, Y = y)$
(2, 1)	0.0625
(3, 2)	0.1250
(4, 2)	0.0625
(4, 3)	0.1250
(5, 3)	0.1250
(5, 4)	0.1250
(6, 3)	0.0625
(6, 4)	0.1250
(7, 4)	0.1250
(8, 4)	0.0625

6. Starting with the two-way table, how could you obtain the marginal distribution of X ? of Y ?

7. Starting with the marginal distribution of X and the marginal distribution of Y , could you necessarily construct the two-way table of the joint distribution? Explain.

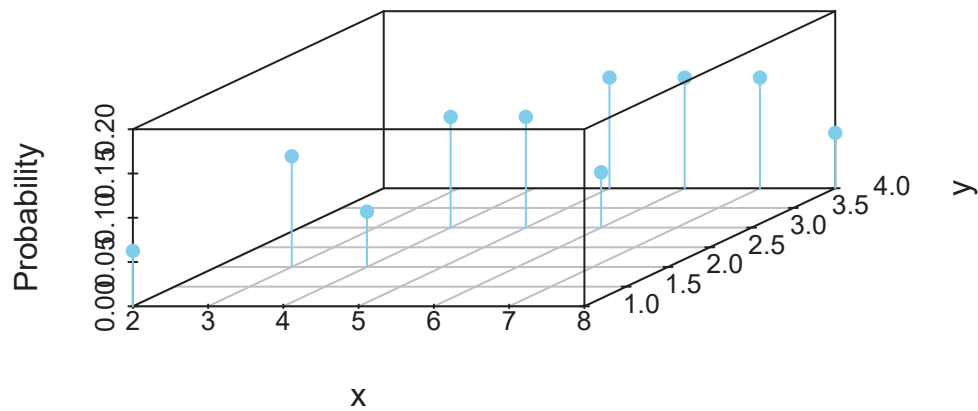
- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) pairs, and describes how the values of X and Y vary together or jointly.
- Marginal distributions can be obtained from a joint distribution by “stacking”/“collapsing”/“aggregating” out the other variable.
- **In general, marginal distributions alone are not enough to determine a joint distribution.** (The exception is when random variables are *independent*.)

Table 12.2: Two-way table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0

3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

3D Impulse Plot



Example 12.2. Continuing the dice rolling example, construct a spinner representing the joint distribution of X and Y .

- The **joint probability mass function (pmf)** of two *discrete* random variables (X, Y) defined on a probability space is the function defined by

$$p_{X,Y}(x, y) = P(X = x, Y = y) \quad \text{for all } x, y$$

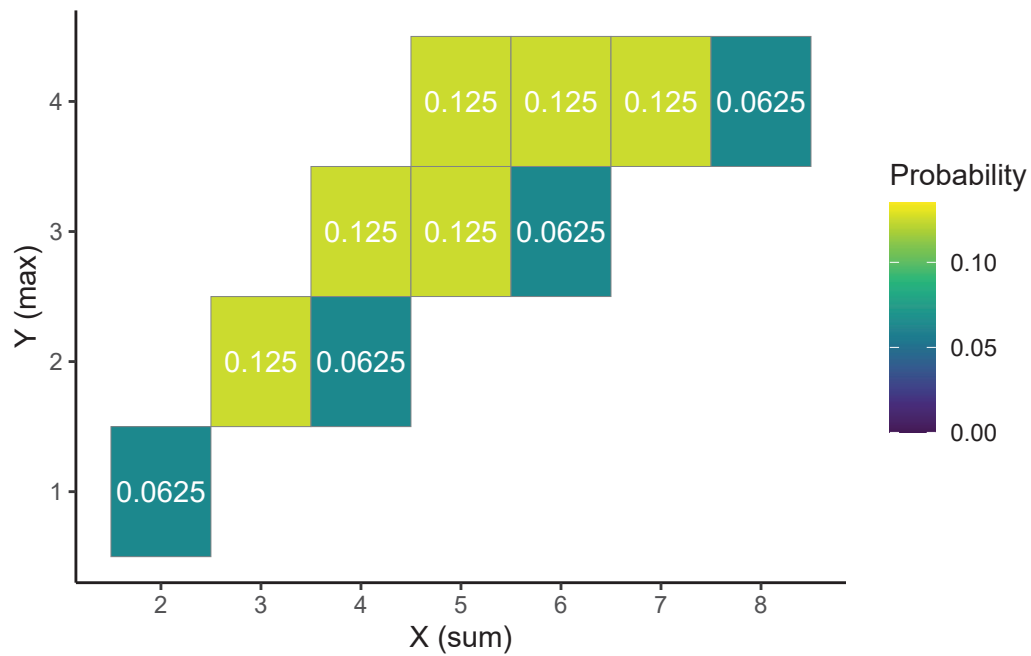


Figure 12.1: Tile plot representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

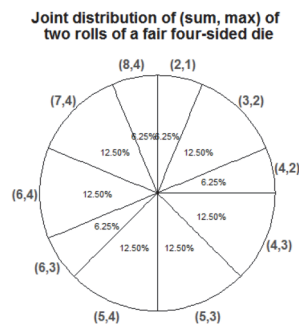


Figure 12.2: Spinner representing the joint distribution of the sum (X) and larger (Y) of two rolls of a fair four-sided die.

- Remember to specify the possible (x, y) pairs when defining a joint pmf.

Example 12.3. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y have joint pmf

$$p_{X,Y}(x, y) = \begin{cases} e^{-2.3} \frac{1.2^x 1.1^y}{x!y!}, & x = 0, 1, 2, \dots; y = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

1. Compute and interpret the probability that the home teams hits 2 home runs and the away team hits 1 home run.
2. Construct a two-way table representation of the joint pmf (you can use software or a spreadsheet).
3. Compute and interpret the probability that each team hits at most 3 home runs.
4. Compute and interpret the probability that both teams combine to hit a total of 3 home runs.
5. Compute and interpret the probability that the home team and the away team hit the same number of home runs.

6. Compute and interpret the probability that the home team hits 2 home runs.

7. Compute and interpret the probability that the away team hits 1 home run.

- Recall that we can obtain marginal distributions from a joint distribution.
- *Marginal pmfs* are determined by the joint pmf via the law of total probability.
- If we imagine a plot with blocks whose heights represent the joint probabilities, the marginal probability of a particular value of one variable can be obtained by “stacking” all the blocks corresponding to that value.

$$p_X(x) = \sum_y p_{X,Y}(x, y) \quad \text{a function of } x \text{ only}$$
$$p_Y(y) = \sum_x p_{X,Y}(x, y) \quad \text{a function of } y \text{ only}$$

12.1 Joint probability density functions

- The joint distribution of two continuous random variables can be specified by a joint pdf, a *surface* specifying the density of (x, y) pairs.
- The probability that the (X, Y) pair of random variables lies in some region is the *volume* under the joint pdf surface over the region.
- The **joint probability density function (pdf)** of two *continuous* random variables (X, Y) defined on a probability space is the function $f_{X,Y}$ which satisfies, for any region

S

$$P[(X, Y) \in S] = \iint_S f_{X,Y}(x, y) dx dy$$

- A joint pdf is a surface with height $f_{X,Y}(x, y)$ at (x, y) .
- The probability that the (X, Y) pair of random variables lies in the region A is the *volume* under the pdf surface over the region A
- The height of the density surface at a particular (x, y) pair is related to the probability that (X, Y) takes a value “close to” (x, y) :

$$P(x - \epsilon/2 < X < x + \epsilon/2, y - \epsilon/2 < Y < y + \epsilon/2) = \epsilon^2 f_{X,Y}(x, y) \quad \text{for small } \epsilon$$

- The joint distribution is a distribution on (X, Y) pairs. A mathematical expression of a joint distribution is a function of both values of X and values of Y . Pay special attention to the possible values; the possible values of one variable might be restricted by the value of the other.
- The marginal distribution of Y is a distribution on Y values only, regardless of the value of X . A mathematical expression of a marginal distribution will have only values of the single variable in it; for example, an expression for the marginal distribution of Y will only have y in it (no x , not even in the possible values).

12.2 Independence of random variables

- Two random variables X and Y defined on a probability space with probability measure P are **independent** if $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$ for all x, y . That is, two random variables are independent if their joint cdf is the product of their marginal cdfs.
- Random variables X and Y are independent if and only if the joint distribution factors into the product of the marginal distributions. The definition is in terms of cdfs, but analogous statements are true for pmfs and pdfs.

Discrete RVs X and Y are independent

$$\iff p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y$$

Continuous RVs X and Y are independent

$$\iff f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y$$

- Random variables X and Y are independent if and only if their joint distribution can be factored into the product of a function of values of X alone and a function of values of Y alone. That is, X and Y are independent if and only if there exist functions g and h for which

$$f_{X,Y}(x, y) \propto g(x)h(y) \quad \text{for all } x, y$$

- X and Y are independent if and only if the joint distribution factors into a product of the marginal distributions. The above result says that you can determine if that's true without first finding the marginal distributions.

Example 12.4. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y are independent, X has a Poisson(1.2) distribution and Y has a Poisson(1.1) distribution.

1. Find the joint pmf of X and Y .
2. How could you have from the setup of Example 12.3 that X and Y were independent?